

Training-Free Stability Components for Cross-Backend Qubit Quality Prediction on IBM Quantum Hardware

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Abstract

We investigate whether the three normalized components of a thermodynamic stability metric, normalized fidelity I , coherence ratio $\rho = T_2/T_1$, and readout error S , can serve as training-free input features for machine learning models that predict the Φ -threshold qubit-quality class across IBM Quantum hardware. Using a strict methodology that excludes Φ from the feature set and evaluates generalization via backend-split evaluation, we find that on 445 qubits from three IBM Quantum backends, (I, ρ, S) predict the Φ -threshold class with 98.4% average balanced accuracy on held-out backends without retraining. Feature ablation reveals that ρ alone achieves 94.6% balanced accuracy and carries 70–78% of feature importance, while I alone and S alone are near random at 52.2% and 50.9% respectively. All four tested model architectures (Random Forest, Gradient Boosting, Neural Network, SVM) exceed 80% balanced accuracy under the strict protocol. For the narrower task of predicting T2-based hardware quality, a data-driven threshold of 0.60 outperforms the physics-derived threshold of 0.25 by 16.2 percentage points. These results establish (I, ρ, S) as a model-agnostic, training-free feature representation for predicting the Φ -threshold qubit-quality class across the tested IBM Quantum backends, with coherence ratio ρ carrying most of the predictive signal.

1 Introduction

Quantum computing systems require accurate assessments of qubit quality before circuit execution. Poor qubit selection increases circuit error rates, wastes compute resources, and in multi-qubit systems can propagate errors across the entire computation. Current approaches to qubit quality monitoring are primarily reactive or platform-specific, relying on calibration metrics and heuristics that do not obviously transfer across hardware backends without revalidation (IBM, 2025).

Prior work on qubit selection has focused on heuristic cost functions. Nation and Treinish (Nation and Treinish, 2023) developed mapomatic, a post-compilation layout optimizer that multiplies individual error rates to score qubit subsets. Their default cost function explicitly excluded T_1/T_2 -derived idle-time error because it did not materially affect layout ordering in their experiments. In the backend-split stability-classification setting studied here, the coherence ratio $\rho = T_2/T_1$ carries most of the predictive signal, accounting for 70–78% of feature importance in the strict evaluation.

The stability metric at the center of this work is:

$$\Phi = I \times \rho - \alpha \times S \quad (1)$$

where $I = (F - 0.50)/0.50$ is normalized gate fidelity above the random baseline for two-level systems, $\rho = \min(T_2/T_1, 1.0)$ is the coherence ratio, S is readout error, and $\alpha = 0.1$ is a coupling constant. The threshold $\Phi_c = 0.25$ classifies qubits as GOOD ($\Phi \geq 0.25$), MARGINAL ($0 \leq \Phi < 0.25$), or BAD ($\Phi < 0$). Prior work using the same Φ formulation validated the metric on real IBM Quantum hardware across 445 qubits and three backends (Barnicle, 2026).

This paper addresses a different question: rather than asking whether Φ discriminates qubit quality, we ask whether the individual components (I, ρ, S) can serve as input features to a machine learning model that predicts the Φ -threshold class across unseen hardware backends. This framing avoids a circularity present in earlier exploratory work: if Φ appears in the feature vector while Φ also defines the classification label, the resulting model is learning a tautology rather than a generalizable relationship. The strict tests reported here correct this by removing Φ from features entirely and evaluating generalization through backend-split transfer experiments on 445 real qubits across `ibm_fez`, `ibm_torino`, and `ibm_marrakesh`.

The main contributions of this paper are:

1. A strict backend-split evaluation showing that (I, ρ, S) predict the Φ -threshold class at 98.4% average balanced accuracy across IBM Quantum backends without retraining.
2. A feature ablation establishing that $\rho = T_2/T_1$ is the dominant predictive component, accounting for 70–78% of feature importance, while I and S alone are near random.
3. Evidence that the feature representation is model-agnostic across Random Forest, Gradient Boosting, Neural Network, and SVM architectures.
4. A threshold optimization result showing that for T2-based quality prediction specifically, a data-driven threshold of 0.60 outperforms the physics-derived 0.25 by 16.2 percentage points.

2 Methods

2.1 Data

Calibration data were obtained from three IBM Quantum backends: `ibm_fez` (156 qubits), `ibm_torino` (133 qubits), and `ibm_marrakesh` (156 qubits), yielding 445 analyzed qubits total. Each qubit record contains single-qubit gate fidelity F , relaxation time T_1 , coherence time T_2 , and readout error E . No synthetic data were used at any stage.

2.2 Feature Computation

Three features were computed for each qubit from calibration data:

$$I = \frac{F - 0.50}{1 - 0.50} = 2F - 1 \quad (2)$$

$$\rho = \min\left(\frac{T_2}{T_1}, 1.0\right) \quad (3)$$

$$S = E \quad (4)$$

The composite stability metric is $\Phi = I \times \rho - 0.1 \times S$, and the baseline classification threshold is $\Phi_c = 0.25$. The classification label for each qubit is GOOD if $\Phi \geq 0.25$ and NOT-GOOD otherwise. Φ itself was excluded from the feature vector in all experiments reported here to avoid a circularity in which the label-defining quantity also appears as a predictor.

2.3 Strict Methodology and Circularity Correction

An earlier set of exploratory tests for this repo included Φ in the feature set while using $\Phi \geq 0.25$ as the classification label. Because Φ is a deterministic function of (I, ρ, S) , a model trained with Φ as a feature is learning a tautology rather than a generalizable component relationship. Those exploratory results are documented in the repository for historical context but are not reported here as primary scientific evidence.

The strict tests reported in this paper apply two methodological corrections. First, Φ is excluded from the feature vector. The feature set is (I, ρ, S) only, with $\Phi \geq 0.25$ retained as the label. Second, train/test splitting is done by backend rather than by random qubit assignment. For each evaluation direction, a model was trained on one backend and tested on one held-out backend. Results are then averaged across all cross-backend train/test directions. This backend-split protocol tests whether the component relationship generalizes across hardware rather than within a single calibration snapshot.

Accordingly, the strict classification experiments test whether the components (I, ρ, S) reproduce the disclosed Φ -threshold classification consistently across hardware, not whether machine learning independently validates the physical threshold itself.

Balanced accuracy is used as the primary metric for the binary classification experiments to account for class imbalance, as GOOD qubits constitute 391 of 445 qubits (87.9%) in the tested dataset. Exceptions are noted where other metrics are more appropriate, including accuracy for the pair-prediction experiment and R^2 for regression.

2.4 Machine Learning Models

Four model architectures were evaluated: Random Forest, Gradient Boosting, Neural Network, and Support Vector Machine, all implemented in scikit-learn (Pedregosa et al., 2011). Each model was trained on one IBM Quantum backend and evaluated separately on each of the other two backends. Results are reported as averages across all directional train/test backend pairs. Feature importance was computed from Random Forest and Gradient Boosting models using mean impurity decrease.

2.5 Feature Ablation

To quantify the contribution of each component, models were trained and evaluated under the same strict backend-split protocol using five feature subsets: ρ alone, I alone, S alone, $I + \rho$, and $I + \rho + S$. The same balanced accuracy metric and backend-split evaluation were applied consistently across all subsets.

2.6 Threshold Optimization

To test whether the physics-derived threshold $\Phi_c = 0.25$ is optimal for all downstream tasks, a threshold sweep was conducted for the specific task of predicting T2-based hardware quality. Ground truth labels for this task were defined as $T_2 \geq \text{median } T_2$ across the dataset, independent of Φ . Eleven threshold values were tested. For each threshold θ , training labels were defined by $\Phi \geq \theta$ on the training backend, and predictions were evaluated against the held-out backend’s T2-based ground-truth labels. Balanced accuracy was computed for each threshold value. This analysis tests threshold choice for the specific downstream task of T2-based quality prediction and does not replace the physics-derived threshold $\Phi_c = 0.25$ for the base Φ classification system.

2.7 Multi-Qubit Pair Prediction

To test whether the component features extend to multi-qubit system quality prediction, 442 qubit pairs were constructed from the 445-qubit dataset using the repository’s fixed pair-generation script, applied consistently across all pair-level experiments; the exact construction rule is available in the released code. For each pair, the feature vector consisted of the six concatenated component values $(I_1, \rho_1, S_1, I_2, \rho_2, S_2)$. The label was defined by whether the minimum Φ of the pair satisfied $\min(\Phi_1, \Phi_2) \geq 0.25$. The same backend-split evaluation protocol was applied.

2.8 Multi-Target Regression

To assess whether the component features carry predictive information about hardware metrics beyond the Φ -threshold class, separate regression models were trained to predict T_2 and T_1 from (I, ρ, S) under the same backend-split protocol. R^2 on held-out backends was used as the primary metric. Note that $\rho = T_2/T_1$ means the T_2 regression has partial structural overlap with the feature set; this is reported as a caveat on those results.

2.9 Three-Way Classification

A multiclass experiment used three quality categories: GOOD ($\Phi \geq 0.25$), MARGINAL ($0.15 \leq \Phi < 0.25$), and BAD ($\Phi < 0.15$). This three-way partition is an experiment-specific ML label scheme for the strict multiclass test and should not be confused with the original physics-motivated GOOD/MARGINAL/BAD partition used elsewhere in the Φ framework. The same backend-split evaluation and (I, ρ, S) feature set were used. Both overall accuracy and balanced accuracy are reported because class imbalance is more severe in the three-class setting than in the binary case.

3 Related Work

Qubit quality assessment and selection have been studied primarily through heuristic cost functions and post-compilation routing. Nation and Treinish (Nation and Treinish, 2023) developed mapomatic, which recovers approximately 40% of missing fidelity by routing compiled circuits to low-noise hardware subgraphs. Their cost function multiplies individual error rates and explicitly excludes T_1/T_2 -derived idle-time error, a design choice motivated by the observation that coherence information did not materially affect layout

ordering in their single-snapshot experiments. The present work studies a different context: cross-backend stability classification rather than single-snapshot layout ordering. In that context, the coherence ratio $\rho = T_2/T_1$ accounts for 70–78% of the predictive signal, indicating a substantially larger role for coherence information than is reflected in layout-ordering heuristics.

Transfer learning for quantum hardware has received limited attention in the sources considered here. Pan and Yang (Pan and Yang, 2010) provide a foundational survey of transfer learning methods across domains. The present work studies cross-backend transfer: a model trained on calibration features from one IBM backend is evaluated on a held-out backend with no parameter updates, achieving 98.4% balanced accuracy on average.

Prior machine-learning approaches to qubit quality prediction cited here were evaluated within platform-specific settings rather than under the backend-split transfer protocol used in this paper. Patel et al. (Patel et al., 2020) demonstrated operation-aware error rate prediction for NISQ devices from hardware calibration data.

The stability metric $\Phi = I \times \rho - \alpha \times S$ used in this work was previously validated on real IBM Quantum hardware across 445 qubits in a temporal early-warning study (Barnicle, 2026). That work established Φ as a leading indicator of qubit degradation and demonstrated 83% error reduction through Φ -based qubit selection. The present work addresses a distinct question: whether the individual components (I, ρ, S) carry generalizable cross-backend signal when used as ML input features without Φ itself in the feature set.

4 Results

4.1 Cross-Backend Classification

Under the strict backend-split evaluation with (I, ρ, S) as features and $\Phi \geq 0.25$ as the label, models trained on one IBM Quantum backend achieved 98.4% average balanced accuracy on held-out backends. Within-backend balanced accuracy was 100%, so the average cross-backend result represents only a small performance reduction.

Table 1: Cross-Backend Classification Results (Strict Evaluation, (I, ρ, S) features only)

Train Backend	Test Backend	Balanced Accuracy (%)
ibm_fez	ibm_torino	100.0
ibm_fez	ibm_marrakesh	98.4
ibm_torino	ibm_fez	97.2
ibm_torino	ibm_marrakesh	95.2
ibm_marrakesh	ibm_fez	99.6
ibm_marrakesh	ibm_torino	100.0
Average transfer		98.4

The worst transfer direction was ibm_torino to ibm_marrakesh at 95.2%, indicating modest backend-specific variation but no direction that failed to generalize. These results establish that the (I, ρ, S) component relationship with the Φ -threshold class is stable across the three tested IBM backends.

4.2 Feature Ablation

Feature ablation under the same strict backend-split protocol identified $\rho = T_2/T_1$ as the dominant predictive component.

Table 2: Feature Ablation Results (Strict Backend-Split Evaluation)

Feature Set	Average Balanced Accuracy (%)
ρ alone	94.6
$I + \rho$	98.6
$I + \rho + S$ (full set)	98.4
I alone	52.2
S alone	50.9

ρ alone achieved 94.6% balanced accuracy, while I alone and S alone were near random at 52.2% and 50.9% respectively. $I + \rho$ marginally outperformed the full set (98.6% versus 98.4%), indicating that S did not improve average balanced accuracy in this classification setting. Feature importance computed from Random Forest and Gradient Boosting models consistently assigned 70–78% of importance to ρ , 12–16% to I , and 9–15% to S .

This result is consistent with prior work in this portfolio showing that Φ correlates with T_2/T_1 at $r = 0.9458$ across 445 qubits (Barnicle, 2026). For the stability-classification setting studied here, excluding coherence-ratio information would remove the component carrying 70–78% of the predictive signal, even though coherence information was less useful for the single-snapshot layout-ordering task studied by Nation and Treinish (Nation and Treinish, 2023).

4.3 Model Architecture Comparison

All four tested model architectures achieved above 80% average balanced accuracy under the strict backend-split evaluation.

Table 3: Model Architecture Comparison (Strict Backend-Split Evaluation, (I, ρ, S) features)

Model	Avg BalAcc (%)	Min BalAcc (%)	Max BalAcc (%)
Gradient Boosting	98.7	95.2	100.0
Random Forest	98.4	95.2	100.0
Neural Network	98.1	95.2	100.0
SVM	82.2	52.8	100.0

Tree-based models (Gradient Boosting and Random Forest) performed best and most consistently. SVM achieved 82.2% on average but showed the highest variance across backend directions, with a minimum of 52.8% in one transfer direction. These results establish (I, ρ, S) as a model-agnostic feature representation: the predictive signal generalizes across model families, though tree-based models are recommended for cross-backend deployment.

4.4 Multi-Qubit Pair Prediction

Pair-quality prediction using concatenated component features $(I_1, \rho_1, S_1, I_2, \rho_2, S_2)$ and labels defined by $\min(\Phi_1, \Phi_2) \geq 0.25$ achieved 98.2% average transfer accuracy across backends on 442 qubit pairs. Per-direction results ranged from 96.1% to 100.0%. For the pair experiment, transfer performance is reported as accuracy rather than balanced accuracy, matching the underlying pair-label evaluation used in the strict test outputs.

In the original exploratory tests where Φ was included as a feature, minimum Φ substantially outperformed average Φ (86.5%) and maximum Φ (69.2%) as the pair aggregation rule. Those results are circular with respect to the label definition. The strict transfer result of 98.2% shows that pair-quality prediction based on the weakest-link framing remains stable across backends, although the direct min-versus-average-versus-maximum comparison comes from the earlier exploratory tests.

4.5 Three-Way Classification

Three-way classification using GOOD ($\Phi \geq 0.25$), MARGINAL ($0.15 \leq \Phi < 0.25$), and BAD ($\Phi < 0.15$) labels under the strict backend-split evaluation achieved 98.2% average transfer accuracy and 91.7% average transfer balanced accuracy. These three classes are an operational partition used for the strict multiclass experiment and are not identical to the base physical GOOD/MARGINAL/BAD interpretation used elsewhere in the Φ framework. The gap between accuracy and balanced accuracy reflects class imbalance: GOOD qubits dominate the dataset, making the MARGINAL and BAD classes harder to classify consistently across backends.

4.6 Multi-Target Regression

Under the strict backend-split regression protocol, T_2 prediction from (I, ρ, S) achieved average $R^2 = 0.32$ on held-out backends, while T_1 prediction achieved average $R^2 = -0.28$. The T_1 result indicates no generalizable cross-backend signal for relaxation time from the tested feature set. T_2 carries moderate signal, though the result should be interpreted cautiously because $\rho = T_2/T_1$ means the feature set has partial structural overlap with the T_2 target.

4.7 Threshold Optimization

For the specific task of predicting T2-based hardware quality ($T2 \geq \text{median } T2$ as ground truth), a threshold sweep across eleven values identified 0.60 as the optimal threshold at 78.0% balanced accuracy. The physics-derived threshold of 0.25 achieved 61.8% balanced accuracy on this task, ranking 8th of 11 tested values.

Table 4: Threshold Optimization for T2-Based Quality Prediction (Selected Values)

Threshold	Balanced Accuracy (%)
0.60	78.0 (optimal)
0.55	77.6
0.50	76.1
0.25	61.8 (rank 8 of 11)

The 16.2 percentage point gap between 0.60 and 0.25 on this task establishes that the physics-derived threshold is not optimal for every downstream prediction target. The threshold of 0.25 remains the disclosed baseline for the base Φ classification system. For the narrower task of T2-quality prediction specifically, a data-driven threshold near 0.60 produced materially better results in this study.

5 Discussion

The central finding of this paper is that the three normalized components of the Φ stability metric, taken as a feature vector (I, ρ, S) without Φ itself, predict the Φ -threshold qubit-quality class at 98.4% average balanced accuracy across held-out IBM Quantum backends. The result is non-circular: Φ was excluded from the feature set entirely, and generalization was evaluated through backend-split transfer rather than within-backend random splits.

The dominant component in this prediction is $\rho = T_2/T_1$, which alone achieves 94.6% balanced accuracy and carries 70–78% of feature importance in the tree-based models used for feature-importance analysis. This finding is consistent with earlier work showing that Φ correlates with T_2/T_1 at $r = 0.9458$ across 445 qubits on real IBM hardware (Barnicle, 2026), and it provides a mechanistic explanation for why: the coherence ratio is the primary signal for the cross-backend ML prediction task studied here.

The practical implication is that coherence ratio information is substantially more useful for stability-based qubit quality classification than prior layout-selection heuristics have reflected. Nation and Treinish (Nation and Treinish, 2023) excluded T_1/T_2 -derived information from mapomatic’s default cost function because it did not materially affect single-snapshot layout ordering. The present work does not invalidate that design decision for its original context. It does establish that when the task shifts from layout ordering to cross-backend stability classification, excluding coherence-ratio information removes the component carrying most of the predictive signal in the stability-classification setting studied here.

The threshold optimization result adds a second practical finding. The threshold of 0.25 remains the disclosed baseline for the base Φ classification system. For the narrower task of predicting T2-based hardware quality specifically, a data-driven threshold near 0.60 outperforms 0.25 by 16.2 percentage points in this study. This supports a two-layer interpretation: the base Φ framework provides a physics-motivated stability classification, while a task-specific ML layer can optimize the operating threshold for a particular prediction target.

The model-agnosticism result is encouraging for deployment. All four tested architectures exceed 80% balanced accuracy, and tree-based models perform near 99%. The (I, ρ, S) representation does not depend on a specific ML architecture choice, which reduces the engineering burden for practitioners who want to incorporate these features into existing qubit selection pipelines.

Pair-level prediction achieves 98.2% transfer accuracy using labels defined by the minimum- Φ rule, consistent with the weakest-link interpretation suggested by the exploratory tests. Three-way classification achieves 91.7% balanced accuracy across GOOD, MARGINAL, and BAD classes, establishing that the component features preserve the graded quality structure across backends.

6 Limitations

The results in this paper are bounded to three IBM Quantum backends (ibm_fez, ibm_torino, ibm_marrakesh) and to the 445-qubit dataset used in this study. Whether the 98.4% balanced accuracy and the ρ dominance finding generalize to other IBM backends, to other quantum hardware platforms (Google, IonQ, Rigetti), or to data collected under different hardware conditions is not established by this study.

The classification target throughout this paper is the Φ -threshold class ($\Phi \geq 0.25$), not an externally independent hardware quality measure. The strict tests establish that the component relationship is consistent across backends, not that the Φ threshold itself is independently validated by machine learning. Independent circuit-execution validation of the Φ threshold is provided in prior work (Barnicle, 2026).

The T_2 regression result ($R^2 = 0.32$) should be interpreted cautiously because $\rho = T_2/T_1$ is a direct function of T_2 , creating partial structural overlap between the feature set and the regression target. The T_1 null result ($R^2 = -0.28$) is free of this circularity and is the cleaner of the two regression findings.

The SVM model showed substantially higher variance than tree-based models, with a minimum transfer balanced accuracy of 52.8% in one backend direction. SVM performance exceeded 80% on average but is less reliable for cross-backend deployment than Gradient Boosting or Random Forest.

The threshold optimization result of 0.60 for T2-quality prediction was obtained from the same three-backend dataset used for the classification experiments. An independent holdout dataset for threshold calibration was not available, and the optimal threshold for T2-quality prediction on other hardware or under different calibration conditions may differ.

The three-way classification experiment uses BAD defined as $\Phi < 0.15$, which differs from the physics-motivated BAD definition of $\Phi < 0$ used elsewhere in the Φ framework. This partition was used as an operational ML label scheme for the strict multiclass test and should be interpreted as such.

7 Conclusion

We investigated whether the normalized components of the thermodynamic stability metric Φ , specifically normalized fidelity I , coherence ratio $\rho = T_2/T_1$, and readout error S , carry generalizable predictive signal for qubit quality classification across IBM Quantum hardware when used as training-free input features without Φ in the feature set. Using a strict backend-split evaluation on 445 qubits across three IBM backends, we find:

1. (I, ρ, S) predict the Φ -threshold class at 98.4% average balanced accuracy across held-out backends without retraining.
2. $\rho = T_2/T_1$ is the dominant component, achieving 94.6% balanced accuracy alone and carrying 70–78% of feature importance in tree-based models. I alone and S alone are near random at 52.2% and 50.9%.
3. All four tested model architectures exceed 80% balanced accuracy, showing that the feature representation works across diverse model families, with tree-based models performing best.

4. For T2-based quality prediction specifically, a data-driven threshold of 0.60 outperforms the physics-derived 0.25 by 16.2 percentage points in this study.
5. Pair-quality prediction using concatenated component features achieved 98.2% average transfer accuracy across backends, and three-way GOOD/MARGINAL/BAD classification achieved 91.7% balanced accuracy, extending the binary result to more demanding settings.

These results establish (I, ρ, S) as a model-agnostic, training-free feature representation for predicting the Φ -threshold qubit-quality class across the tested IBM Quantum backends. The coherence ratio ρ carries most of the predictive signal, a finding that has direct implications for qubit selection cost functions that have historically excluded coherence-ratio information.

Code Availability

Code and data are available at: <https://github.com/Wise314/universal-phi-ml>

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IBM Quantum hardware accessed via IBM Quantum Network.

Patent Disclosures

The methods described in this paper are the subject of U.S. Provisional Patent Application No. 63/956,800 (filed January 9, 2026). No license to implement or commercialize the described methods is granted by this publication. All rights reserved.

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